ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

DYNAMICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 2nd year Dynamics paper 1 · practice a.

Written by Academic Central for practice, following engineering dynamics modules at South African universities (e.g. University of Pretoria MSD 210 Dynamics: kinetics of systems of particles, plane kinematics and kinetics of rigid bodies, vibration). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method (for example vector or graphical methods in QUESTION 2) and answers that differ only through rounding. Deduct one mark only once per question for missing units.

QUESTION 1

1.1 (6)

Hanging mass: $10g - T = 10a$ ✓
Block: $T - 0.25 \times 20g = 20a$ ✓
Adding: $98.1 - 49.05 = 30a$ ✓
$a = 1.64$ m/s² ✓
$T = 10(9.81 - 1.635)$ ✓ $= 81.75$ N ✓

1.2 (5)

Energy is conserved (smooth surfaces) ✓
$mgh = \frac{1}{2}kx^2$ ✓
$5 \times 9.81 \times 1.5 = 73.58$ J ✓
$x = \sqrt{(2 \times 73.58 / 2\,000)}$ ✓ $= 0.27$ m ✓

1.3 (3)

Momentum before $= 0 = 60 \times 3 - 120\ v$ ✓✓
$v = 1.5$ m/s (forwards, opposite to the person) ✓

1.4 (3)

The string force passes through the centre, so angular momentum is conserved: $m\ v_1\ r_1 = m\ v_2\ r_2$ ✓✓
$v_2 = 2 \times 0.5 / 0.25 = 4$ m/s ✓

1.5 (3)

$F = \dot{m}\ \Delta v = \rho Q v$ ✓ $= 1\,000 \times 0.02 \times 15$ ✓ $= 300$ N ✓

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QUESTION 2

2.1 (2)

$\omega = 300 \times 2\pi/60 = 31.42$ rad/s ✓
$v_A = \omega \times OA = 31.42 \times 0.1 = 3.14$ m/s, horizontal to the right ✓

2.2 (4)

v_A is horizontal and v_B is horizontal (parallel) ✓; the lines at right angles to them through A and B are parallel and never meet, so the instantaneous centre is at infinity ✓
$\omega_{AB} = 0$ (the rod translates for an instant) ✓
$v_B = v_A = 3.14$ m/s to the right ✓

2.3 (2)

$a_A = \omega^2 \times OA = 31.42^2 \times 0.1$ ✓ $= 98.70$ m/s², directed from A towards O (downwards) ✓

2.4 (5)

B is 0.3873 m to the right of O ($\sqrt{(0.4^2 - 0.1^2)}$), so $r_{B/A} = (0.3873\ i - 0.1\ j)$ m ✓
$a_B = a_A + \alpha \times r_{B/A} - \omega_{AB}^2\ r_{B/A}$, with $\omega_{AB} = 0$ ✓
$a_B\ i = -98.70\ j + \alpha\ k \times (0.3873\ i - 0.1\ j) = -98.70\ j + 0.1\alpha\ i + 0.3873\alpha\ j$ ✓
j components: $0 = -98.70 + 0.3873\alpha$ ✓
$\alpha_{AB} = 254.84$ rad/s² (anticlockwise) ✓

2.5 (3)

i components: $a_B = 0.1 \times 254.84$ ✓ $= 25.48$ m/s² ✓
Directed to the right (away from O) ✓

2.6 $\omega = v/r = 6 / 0.3 = 20$ rad/s ✓✓ (5)
Top point: $2v = 12$ m/s ✓✓
Contact point: 0 m/s ✓

2.7 The point (on the body or its extension) that has zero velocity at that instant; the body rotates about it ✓✓ (4)
Draw lines at right angles to the two known velocities; the instantaneous centre is where they cross ✓✓

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QUESTION 3

3.1 $I = \frac{1}{2}mr^2 = \frac{1}{2} \times 20 \times 0.25^2$ ✓ $= 0.63$ kg·m² ✓ (2)

3.2.1 $I_{end} = mL^2/3 = 3 \times 1.2^2 / 3$ ✓ $= 1.44$ kg·m² ✓ (2)

3.2.2 $I_{end} = I_G + m d^2 = 3 \times 1.2^2/12 + 3 \times 0.6^2$ ✓ $= 0.36 + 1.08 = 1.44$ kg·m² ✓ (2)

3.3 Rod: 1.44 kg·m² ✓ (5)
Disc about its own centre: $\frac{1}{2} \times 2 \times 0.1^2 = 0.01$ kg·m² ✓
Parallel axis: $2 \times 1.3^2 = 3.38$ kg·m² ✓
$I_O = 1.44 + 0.01 + 3.38$ ✓ $= 4.83$ kg·m² ✓

3.4 $k = \sqrt{(I/m)} = \sqrt{(0.625/20)}$ ✓ $= 0.18$ m ✓ (4)
The radius of gyration is the distance from the axis at which the whole mass could be concentrated to give the same moment of inertia ✓✓

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QUESTION 4

4.1.1 $\Sigma M = I\alpha$ ✓ (3)
$10 - 2 = 0.625\alpha$ ✓
$\alpha = 12.8$ rad/s² ✓

4.1.2 $\omega = 1\,500 \times 2\pi/60 = 157.08$ rad/s ✓ $t = 157.08 / 12.8 = 12.27$ s ✓ (2)

4.2.1 $\Sigma M_O = I_O \alpha$: $mg(L/2) = I_O \alpha$ ✓ (3)
$3 \times 9.81 \times 0.6 = 1.44\alpha$ ✓
$\alpha = 12.26$ rad/s² ✓

4.2.2 $a_G = \alpha \times L/2 = 12.26 \times 0.6 = 7.36$ m/s² downwards ✓ (3)
$mg - R = m\,a_G$ ✓
$R = 3(9.81 - 7.36) = 7.36$ N upwards ✓

4.2.3 Loss of potential energy = gain in kinetic energy: $mg(L/2) = \frac{1}{2} I_O \omega^2$ ✓✓ (4)
$17.66 = 0.72\omega^2$ ✓
$\omega = 4.95$ rad/s ✓

4.3.1 Along the slope: $mg \sin\theta - F = ma$; about the centre: $F\,r = \frac{1}{2}mr^2\,\alpha$, with $\alpha = a/r$ ✓ (4)
So $F = \frac{1}{2}ma$ ✓ and $mg \sin\theta = 1.5\,ma$, $a = (2/3)\,g \sin\theta$ ✓
$a = (2/3)(9.81)(\sin 20°) = 2.24$ m/s² ✓

4.3.2 $F = \frac{1}{2}ma = (1/3)\,mg \sin\theta$ and $N = mg \cos\theta$ ✓ (3)
$\mu \geq F/N = (1/3) \tan 20°$ ✓ $= 0.12$ ✓

4.3.3 $v^2 = 2as = 2 \times 2.237 \times 5$ ✓ $= 22.37$ ✓ (3)
$v = 4.73$ m/s ✓

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QUESTION 5

5.1 $\omega n = \sqrt{(k/m)} = \sqrt{(4\ 000/10)}$ ✓ $= 20$ rad/s ✓ (3)
$fn = 20/2\pi = 3.18$ Hz ✓

5.2 $c_c = 2\sqrt{(km)} = 2\sqrt{(40\ 000)} = 400$ N·s/m ✓ (4)
$\zeta = 80/400 = 0.2$ ✓
$\omega d = \omega n\sqrt{(1 - \zeta^2)} = 20\sqrt{0.96}$ ✓ $= 19.60$ rad/s ✓

5.3 $\delta = 2\pi\zeta / \sqrt{(1 - \zeta^2)}$ ✓ $= 2\pi \times 0.2 / 0.9798$ ✓ $= 1.28$ ✓ (3)

5.4.1 $r = \omega/\omega n = 0.5$ and $F_0/k = 0.025$ m ✓ (3)
$X = 0.025 / \sqrt{((1 - 0.25)^2 + (2 \times 0.2 \times 0.5)^2)}$ ✓
$= 0.025 / 0.7762 = 0.03$ m (32.21 mm) ✓

5.4.2 At $r = 1$: $X = (F_0/k) / (2\zeta)$ ✓ $= 0.025 / 0.4 = 0.06$ m (62.5 mm) ✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE